

Marine Litter Hazard Assessment in the Sardinian Sea

HHE Data Science Laboratory · Politecnico di Milano

MSFD Descriptor 10 · ISPRA Data 2020–2023

6 BEACHES

41 surveys

14 STATIONS

710 observations

4 YEARS

2020–2023

Research Question

How is marine litter hazard spatially distributed along the Sardinian coast, and what temporal patterns emerge from ISPRA monitoring data (2020–2023)?

WHY IT MATTERS

Mediterranean: 7% of global marine plastic, <1% of ocean surface

Sardinia: 1,849km coastline
3.5M tourists/year
Fragile marine ecosystems (Posidonia, coralligenous)

KNOWLEDGE GAP

No unified spatial hazard map for Sardinian coast

ISPRA data collected since 2015 but never spatially analyzed at sub-regional scale

OUR APPROACH

IPCC hazard framework at sub-regional scale

IDW interpolation on 10km grid

2 compartments:
beach + floating litter



01 — DATA

Sources · Structure · Cleaning

ISPRA MSFD Monitoring Data

MODULO 4 · Beach Litter

6

beaches

41

surveys

2020–2023

years

- 2 surveys/year (spring + autumn)
- Metric: items / 100m transect
- EU JointListCategory standard
- 6 sites along Sardinian coast

MODULO 2BIS · Floating Litter

14

stations

710

observations

2021–2023

years

- Bimonthly surveys
- Metric: observations per station
- Ship-based visual transect
- Offshore monitoring network

Data Harmonization

Schema differences between 2020 and 2023 file formats

Issue	2020 Format	2023 Format	Solution
Survey ID	SampleID	CodiceCampionamento	→ survey_id
Item count	NumeroItems	NumeroOggetti	→ n_items
Transect length	Lunghezza	LunghezzaTransetto	→ transect_m
Material names	Italian (Polimeri artificiali)	English (Artificial polymer)	→ normalized EN

- ✓ All files: transect_m = 100m (fixed MSFD protocol)
- ✓ Missing lat/lon: filled from known station coordinates per beach_id

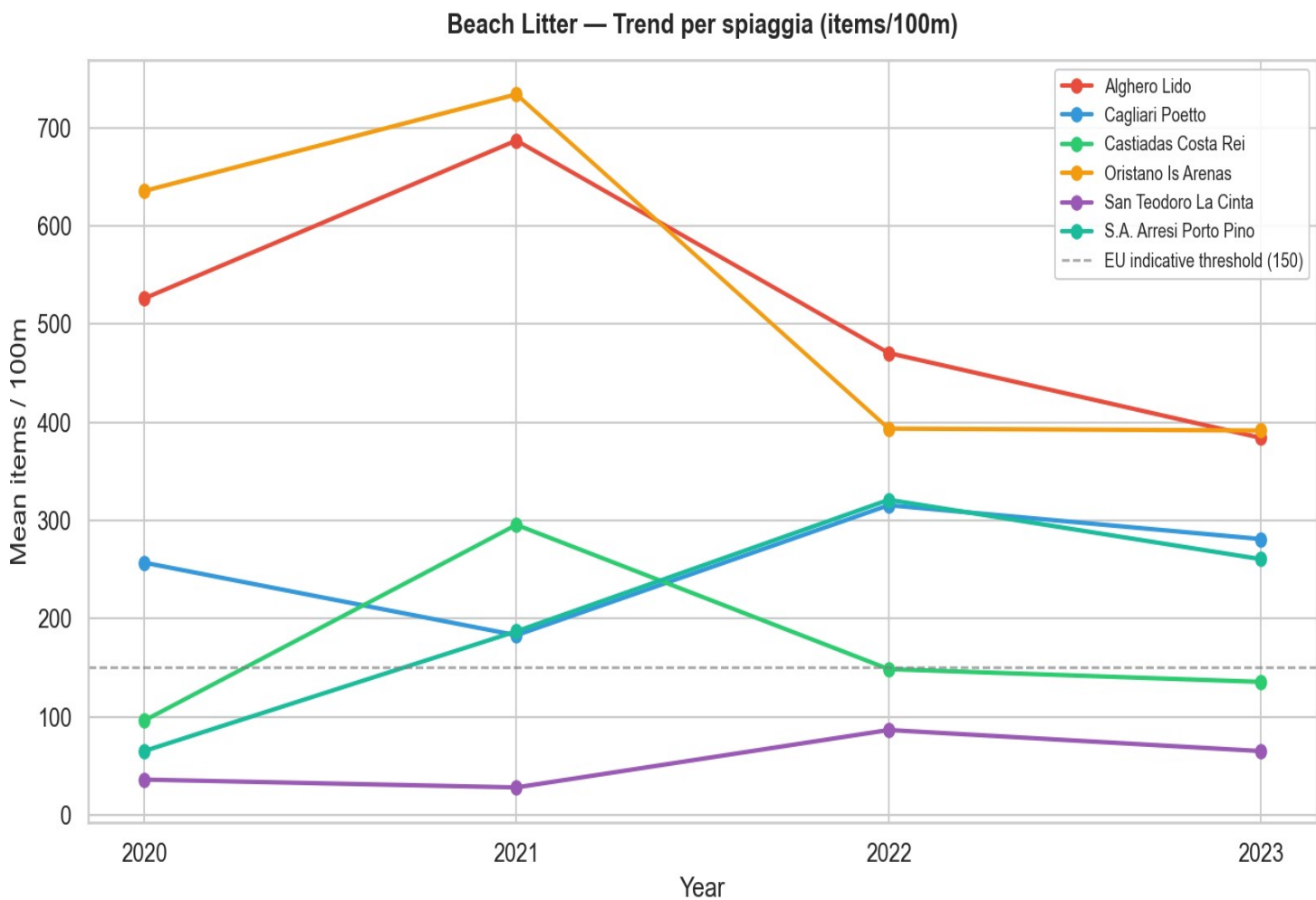




02 — EXPLORATORY ANALYSIS

Beach litter · Floating litter · Patterns

Beach Litter Trend per Stazione (2020-2023)

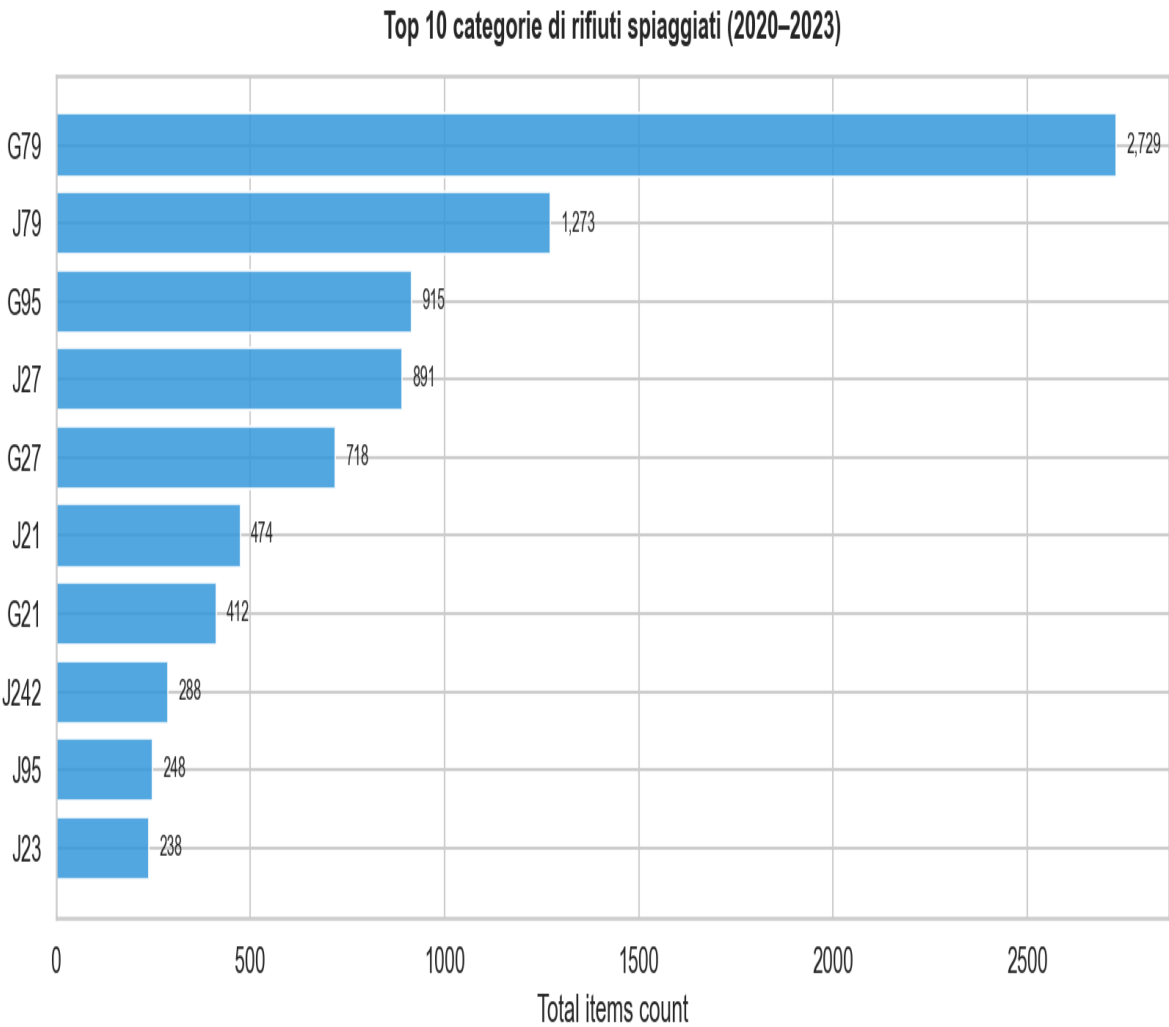


Key Findings

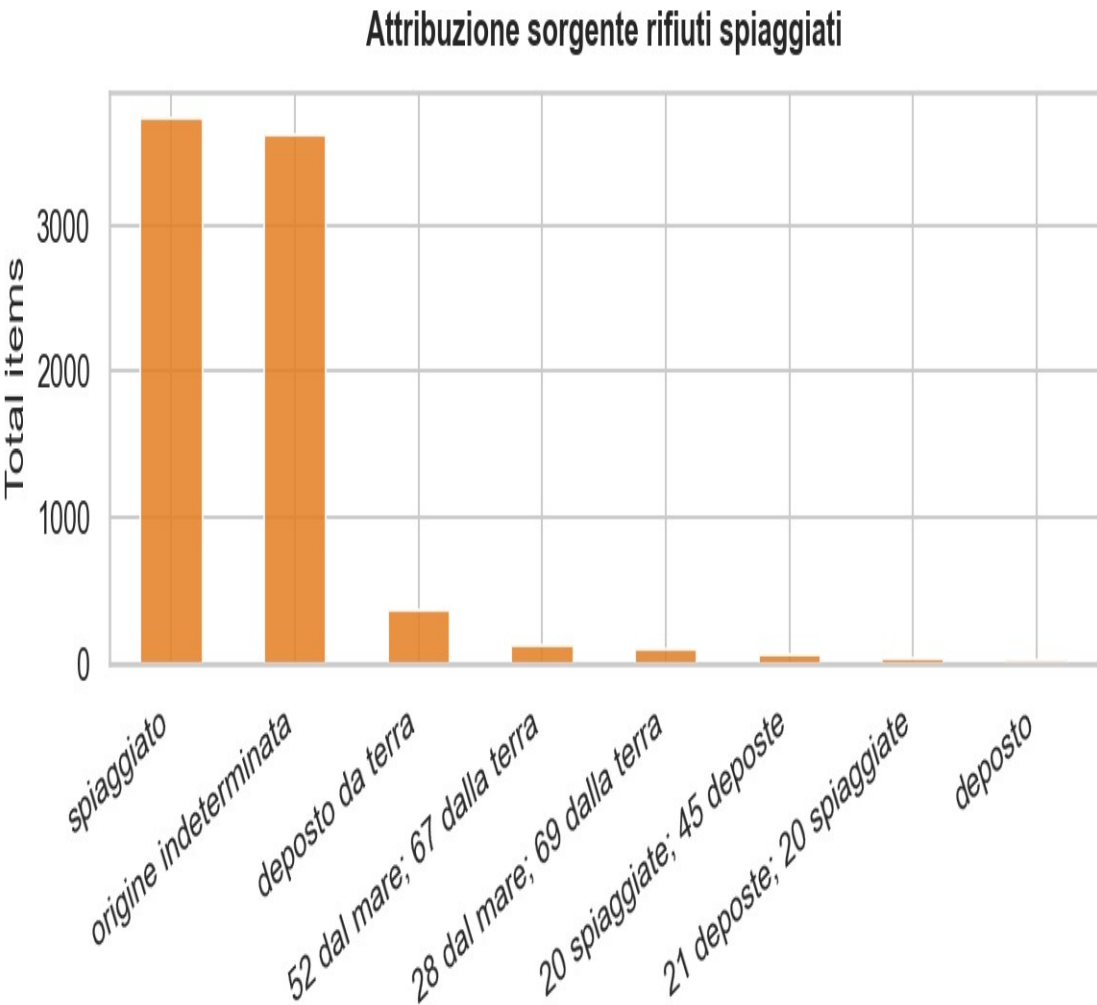
- ▶ Oristano: 539 items/100m (highest)
- ▶ Alghero: 508 items/100m
- ▶ S.Teodoro: 58 items/100m (lowest)
- ▶ EU threshold: 150 items/100m
- ▶ 4/6 beaches exceed threshold
- ▶ West coast > East coast

⚠ West coast significantly above EU threshold

Composition & Source Attribution



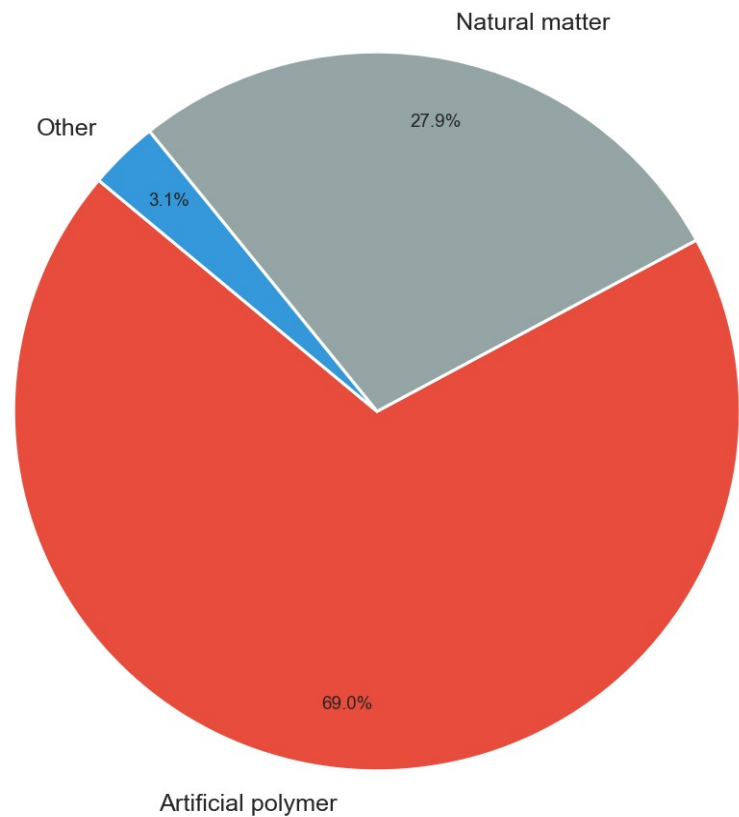
Top 10 categories (JointListCategory)



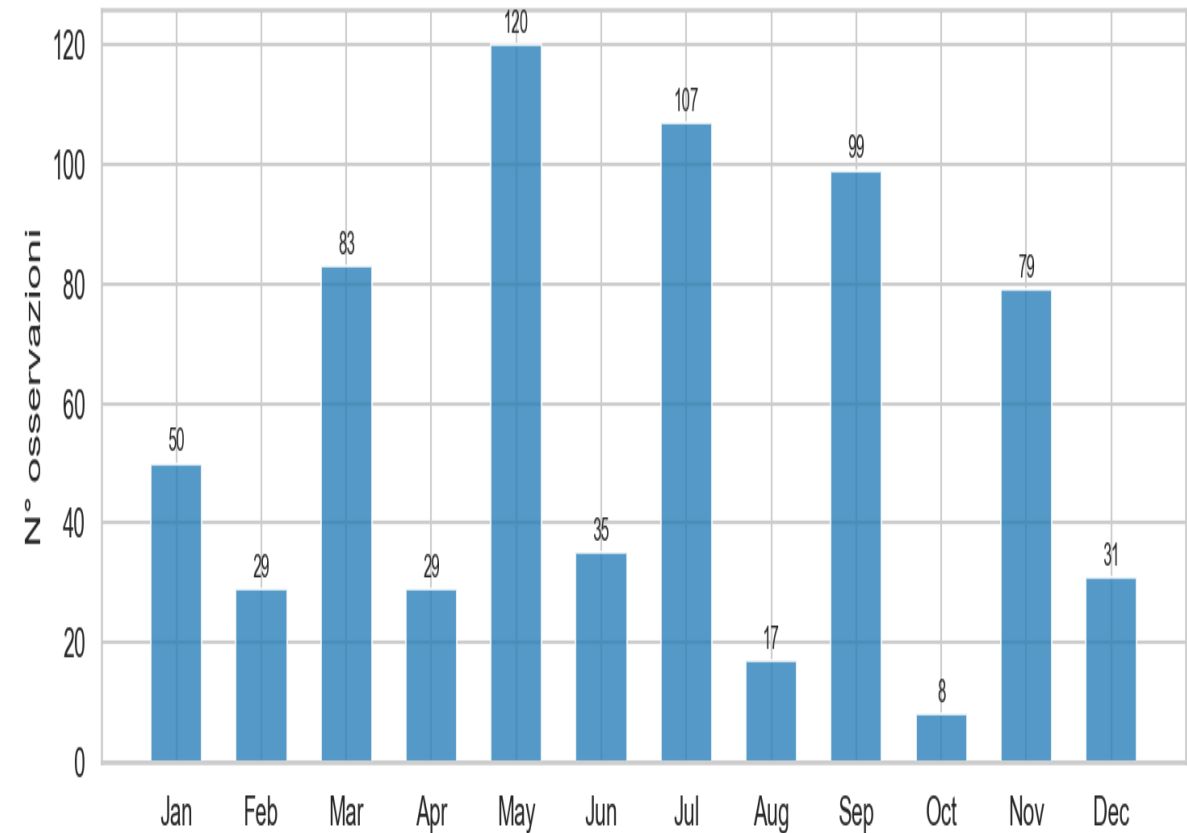
Source attribution (land / sea / fishing)

Floating Litter — Composition & Seasonality

Composizione materiali — rifiuti flottanti (2021–2023)



Stagionalità rifiuti flottanti — osservazioni per mese (2021–2023)



69% of floating litter is artificial polymer (plastic). Peak obs: autumn/winter — consistent with seasonal wind/current patterns.



03 — SPATIAL HAZARD INDEX

Methodology · Results · Hotspots

Hazard Index — Methodology

1

NORMALIZE

Min-max score 0–1 per station.
Global normalization across all years.

2

INTERPOLATE

IDW on 10km grid. Radius 1.0°
beach, 1.2° floating. 289 grid cells.

3

COMPOSITE

Hazard = mean(beach, floating) per
cell. Equal weights.

4

CLASSIFY

5 classes from Very Low to Very
High.

Very Low <0.2

Low 0.2–0.4

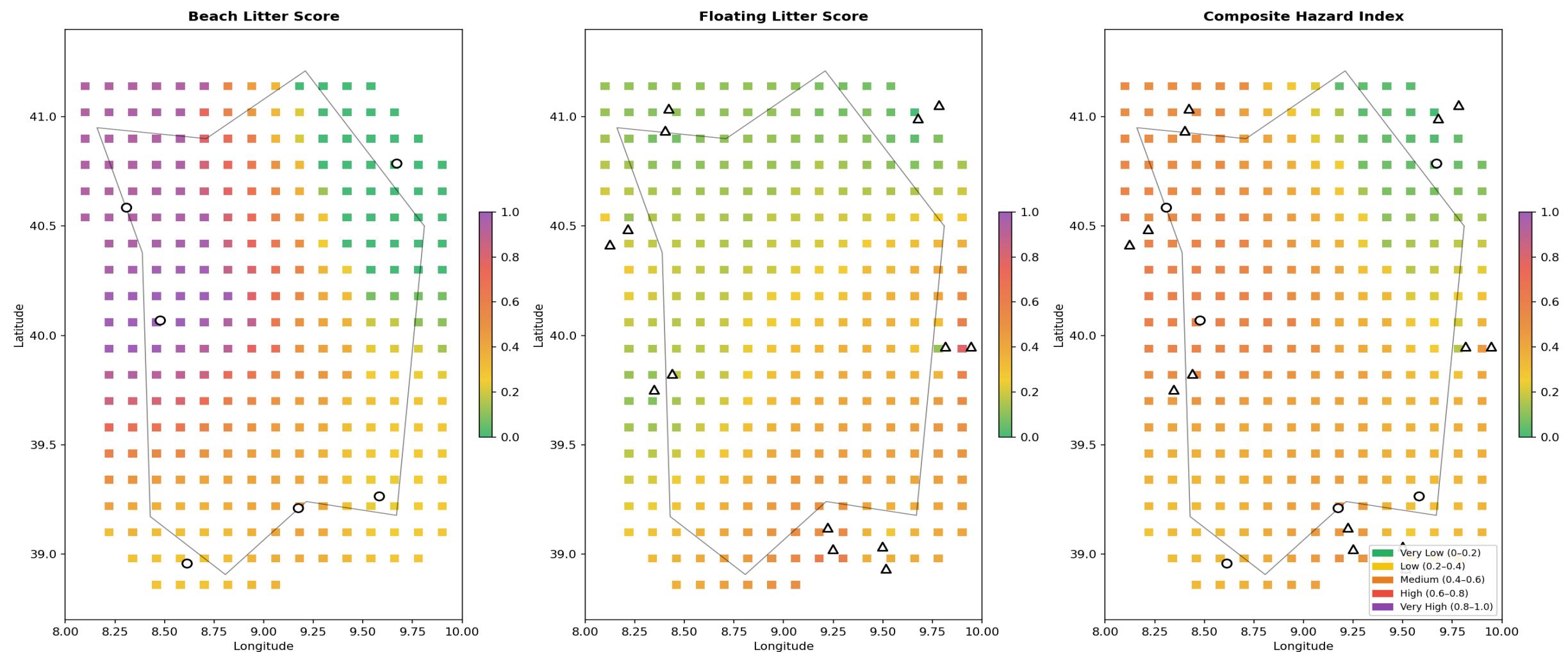
Medium 0.4–0.6

High 0.6–0.8

V.High >0.8

Composite Hazard Index — Sardinia 2020–2023

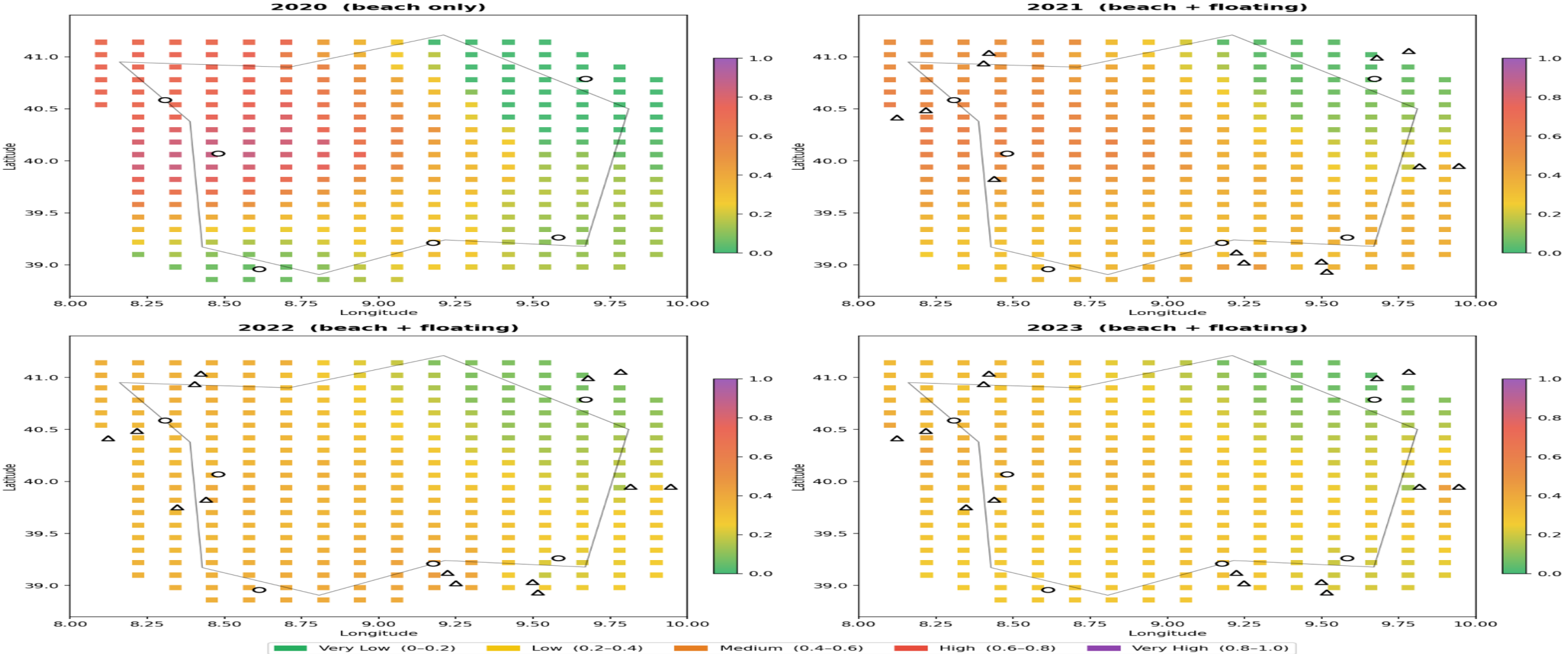
Marine Litter Hazard Index — Sardinia (2020–2023)



Left: beach litter score · Center: floating litter score · Right: composite hazard index | ● beach stations · ▲ floating stations

Temporal Evolution — Hazard Index 2020–2023

Marine Litter Hazard Index — Temporal Evolution (Sardinia 2020–2023)



△ Globally normalized scores — colors comparable across years. 2020: beach compartment only. 2021–2023: beach + floating composite.

Key Findings

HOTSPOTS

- West coast (Alghero–Oristano): highest hazard
- Oristano Is Arenas: beach score 1.00
- Alghero Lido: beach score 0.94
- San Teodoro La Cinta: score 0.00 (lowest)
- 3 grid cells: HIGH class (hazard >0.6)

539

items/100m (Oristano peak)

PATTERNS

- 59% floating litter = artificial polymer
- Beach litter peaks: west coast year-round
- Floating peaks: autumn/winter
- 4/6 beaches exceed EU 150 items/100m threshold
- Trend: apparent reduction 2020→2021 partly methodological

4/6

beaches above EU threshold

Limitations & Next Steps

⚠ LIMITATIONS

- 6 beach stations → sparse, IDW smooths aggressively
- Seafloor compartment missing (no MSFD module locally)
- Score is relative, not absolute scale
- 2020 beach-only ≠ 2021–23 composite
- obs count ≠ standardized density

→ NEXT STEPS

- Retrieve seafloor data (MEDITS) from ISPRA DB
- Add 3rd compartment → complete hazard index
- Replace grid with coastal administrative segments
- Validate vs citizen science beach cleanup data
- Final presentation + policy implications (Phase 4)

Thank You

GitHub: [daniele22-u/hhe-labsardinia](https://github.com/daniele22-u/hhe-labsardinia)

Data: ISPRA — Sistema Nazionale per la Protezione dell'Ambiente

Framework: EU Marine Strategy Framework Directive (MSFD) · Descriptor 10

HHE Data Science Laboratory · Politecnico di Milano · 2026